

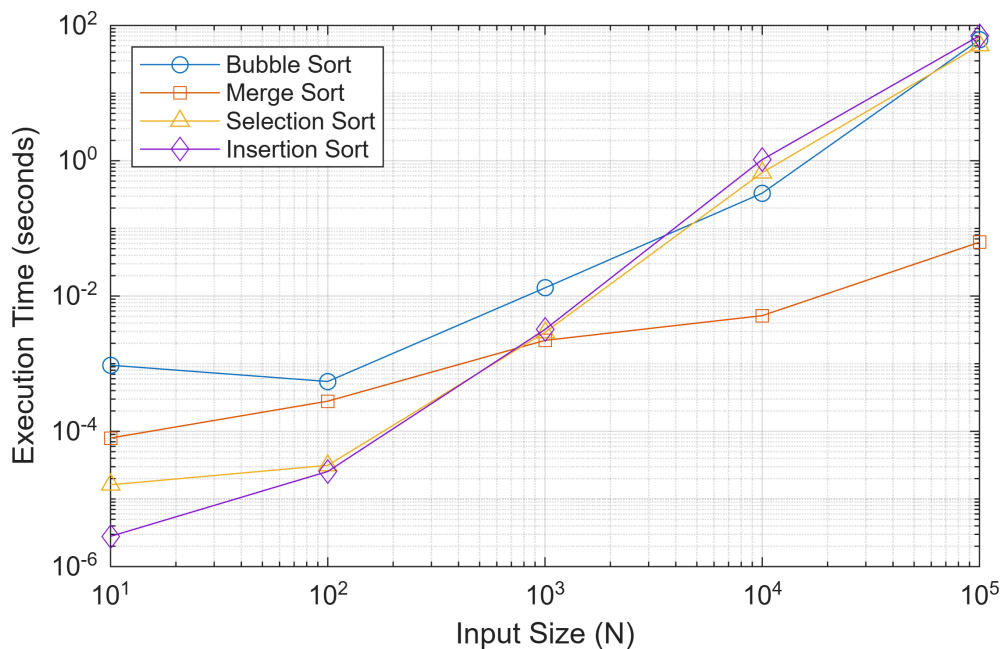
```

bubbleSort = [9.491E-4, 5.456E-4, 0.0132122, 0.3319656, 61.2527651];
mergeSort = [7.96E-5, 2.792E-4, 0.0022101, 0.0051297, 0.0624832];
selectionSort = [1.62E-5, 3.14E-5, 0.0029096, 0.6690401, 50.4041695];
insertionSort = [2.8E-6, 2.55E-5, 0.0032072, 1.0406621, 70.4065892];
size = [10, 100, 1000, 10000, 100000];

figure();
loglog(size, bubbleSort, '-o', ...
        size, mergeSort, '-s', ...
        size, selectionSort, '-^', ...
        size, insertionSort, '-d');

xlabel("Input Size (N)");
ylabel("Execution Time (seconds)");
legend("Bubble Sort", "Merge Sort", "Selection Sort", "Insertion Sort", "Location",
"Northwest");
grid on;

```



Write-up:

With the exception of merge sort, all of the sorting algorithms have a time complexity of  $O(n^2)$ . Merge sort has the shortest time complexity with just  $O(n\log(n))$ . The run times for each sorting algorithm were tested by measuring the run time of each algorithm given a different input size (list size). The lists were generated to be filled with random integers, and each algorithm was given the same lists to ensure fairness. In order to gain a good perspective on how the run times changed with input size, the algorithms were tested multiple times with different input sizes that increased by magnitudes of 10. As seen from the graph depicting the results, for smaller input sizes, ( $n < 10^3$ ), there isn't a significant difference in run time. However, once input sizes started

to grow, merge sort grew at a much smaller rate, an effect of its smaller run time compared to the other sorting algorithms.

Practice Problems:

1.  $T(n) = \Theta(n^2)$
2.  $T(n) = \Theta(n^{\log_2(7)})$
3.  $T(n) = \Theta(n^2 \log_2(n))$
4.  $T(n) = \Theta(n \log_2(n))$
5.  $T(n) = \Theta(n^2)$
6. N/A
7.  $T(n) = \Theta(n^2 (\log_2(n))^2)$
8.  $T(n) = \Theta(n^{\log_2(5)})$
9. N/A
10.  $T(n) = \Theta(\sqrt{n})$
11.  $T(n) = \Theta((\log_2(n))^2)$
12. N/A
13.  $T(n) = \Theta(\sqrt{n})$
14.  $T(n) = \Theta(n (\log_2(n))^2)$
15. N/A
16.  $T(n) = \Theta(\log_2(n) \sqrt{n})$
17.  $T(n) = \Theta(n^{0.51})$
18.  $T(n) = \Theta(n!)$
19.  $T(n) = \Theta(n^{\log_2(3)})$
20.  $T(n) = \Theta(n^2)$
21.  $T(n) = \Theta(n \log_2(n))$
22.  $T(n) = \Theta(n^2)$
23.  $T(n) = \Theta(n^2)$
24.  $T(n) = \Theta(2^n)$
25.  $T(n) = \Theta(n^2)$